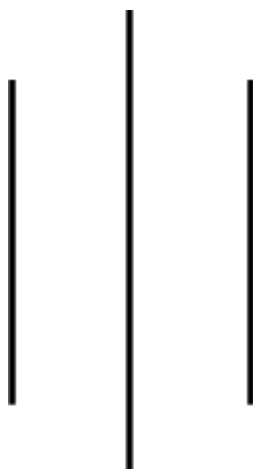




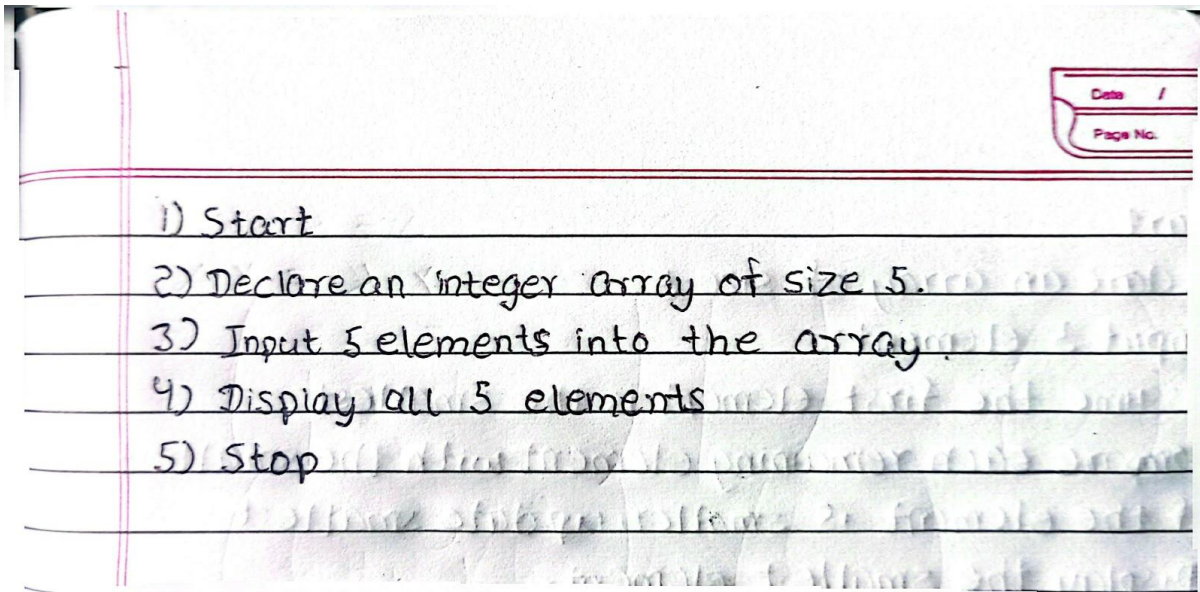
LA GRANDEE INTERNATIONAL COLLEGE

Simalchaur, Pokhara Nepal



Name:	Course	Semester	P.U. Registration Number
Prajwal Sapkota	BCA	3	2025-1-53-0337

1. Write a C program to **input and display 5 elements** of an array.



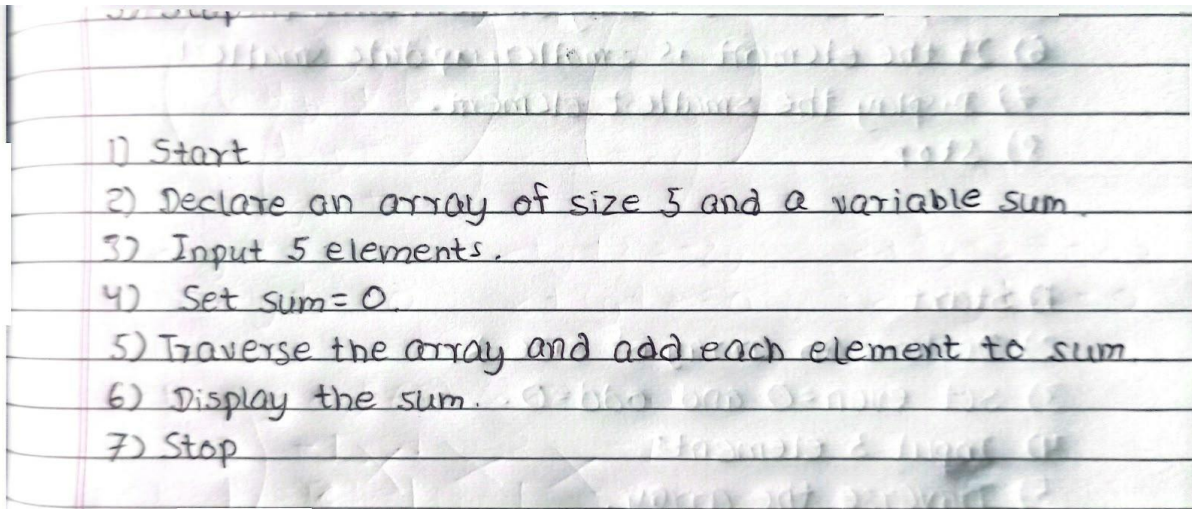
```
#include <stdio.h>
void main()
{
    int arr[5], i;
    printf("Name: Prajwal\n");
    printf("Enter 5 elements:\n");
    for(i = 0; i < 5; i++)
    {
        scanf("%d", &arr[i]);
    }
    printf("The elements are:\n");
    for(i = 0; i < 5; i++)
    {
        printf("%d ", arr[i]);
    }
}
```

```
Name: Prajwal
Enter 5 elements:
5
6
7 8 9
The elements are:
5 6 7 8 9
```

Time Complexity = $O(n)$

Space Complexity = $O(n)$

2. Write a C program to **find the sum of all elements** in an array.



```
#include <stdio.h>
void main()
{
    int arr[5], i, sum=0;
    printf("Enter 5 elements:\n");
    for(i = 0; i < 5; i++)
    {
        scanf("%d", &arr[i]);
    }
    for(i = 0; i < 5; i++)
    {
        sum+=arr[i];
    }
    printf("Sum = %d", sum);
}
```

```
Enter 5 elements:
3
5
6
23
456
Sum = 493
```

Time Complexity = $O(n)$

Space Complexity = $O(n)$

3. Write a C program to **find the largest element** in an array.

- 1) Start
- 2) Declare an array of size 5.
- 3) Input 5 elements.
- 4) Assume the first element is the largest.
- 5) Compare each remaining element with the largest.
- 6) If an element is greater, update largest.
- 7) Display the largest element.
- 8) Stop.

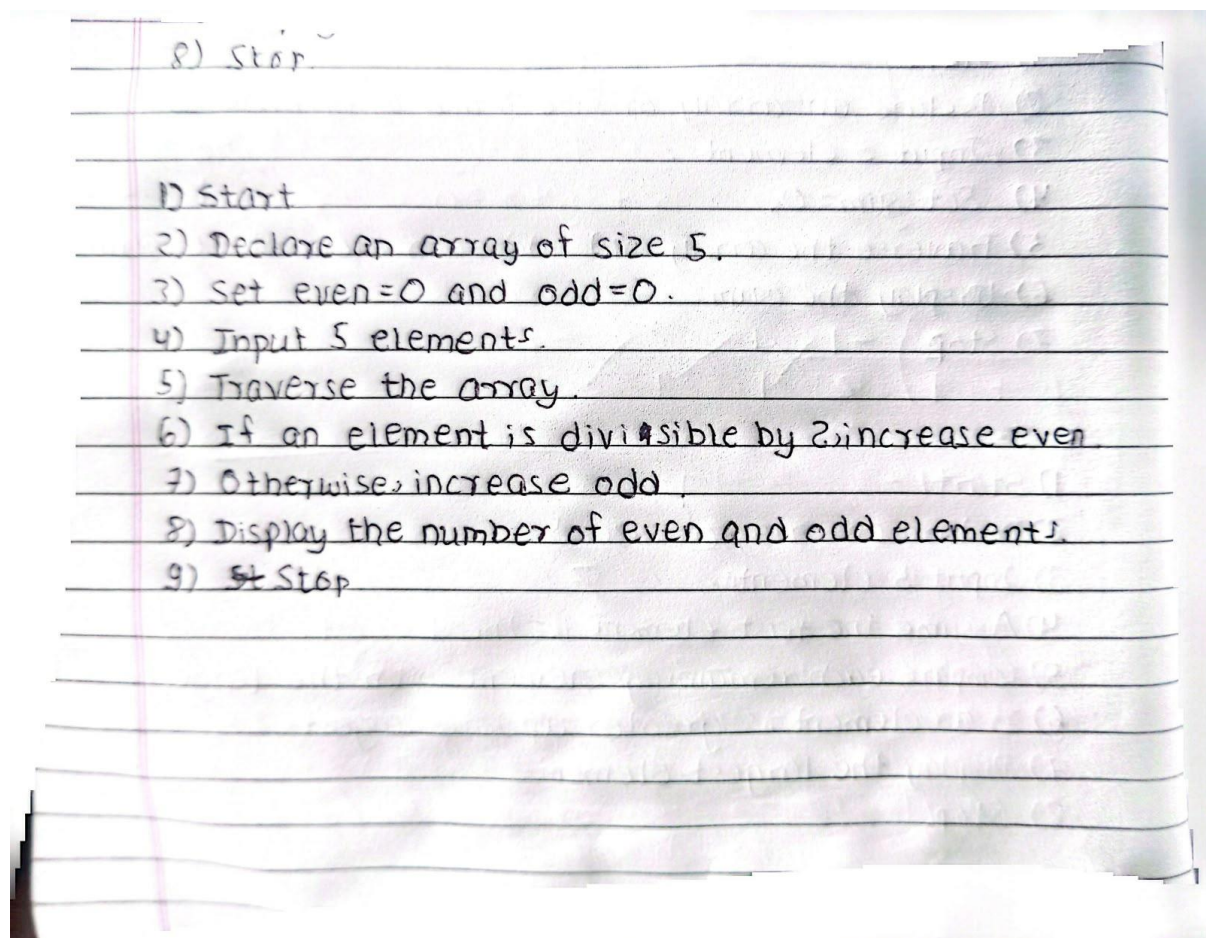
```
#include <stdio.h>
void main()
{
    int arr[5], i, largest;
    printf("Enter 5 elements:\n");
    for(i = 0; i < 5; i++)
    {
        scanf("%d", &arr[i]);
    }
    largest = arr[0];
    for(i = 1; i < 5; i++)
    {
        if(arr[i] > largest)
        {
            largest = arr[i];
        }
    }
    printf("Largest element = %d", largest);
}
```

```
Enter 5 elements:  
5  
8  
44  
67  
78  
Largest element = 78
```

Time Complexity = $O(n)$

Space Complexity = $O(n)$

4. Write a C program to **find the smallest element** in an array.



```
#include <stdio.h>
void main()
{
    int arr[5], i, smallest;
    printf("Enter 5 elements:\n");
    for(i = 0; i < 5; i++)
    {
        scanf("%d", &arr[i]);
    }
    smallest = arr[0];
    for(i = 1; i < 5; i++)
    {
        if(arr[i] < smallest)
        {
            smallest = arr[i];
        }
    }
    printf("Smallest element = %d", smallest);
}
```

```
Enter 5 elements:  
2  
7  
0  
23  
445  
Smallest element = 0
```

Time Complexity = $O(n)$

Space Complexity = $O(n)$

5. Write a C program to **count the number of even and odd elements** in an array.

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- 1) Start
- 2) Declare an array of size 5.
- 3) Input 5 elements.
- 4) Assume the first element is the smallest.
- 5) Compare each remaining element with the smallest
- 6) If the element is smaller, update smallest
- 7) Display the smallest element.
- 8) Stop.

```

#include <stdio.h>
void main()
{
    int arr[5],i, even = 0, odd = 0;
    printf("Enter 5 elements:\n");
    for(i = 0; i < 5; i++)
    {
        scanf("%d", &arr[i]);
    }
    for(i = 0; i < 5; i++)
    {
        if(arr[i] % 2 == 0)
        {
            even++;
        }
        else
        {
            odd++;
        }
    }
    printf("Number of even elements = %d\n", even);
    printf("Number of odd elements = %d", odd);
}

```

```

Enter 5 elements:
4
8
5
78
23
Number of even elements = 3
Number of odd elements = 2

```

Time Complexity = $O(n)$

Space Complexity = $O(n)$

Github : <https://github.com/Cozyanimefan/DSA.git>